Chapter 11. Time Series Analysis

Trend, moving averages, exponential smoothing

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11.1 Types of time series data

Motivation. Data recorded yearly, quarterly or monthly are ordered in time and are usually not independent. Before choosing a method we must recognise which components (trend, seasonality, randomness) a series contains.

A **time series** is a sequence of data observed in time order. It is used to find past patterns, forecast the future, or understand dynamic relationships. A **continuous time series** is observable at every instant and consists of values measured over continuous time. A **discrete time series** is observed only at particular times (yearly, monthly, daily, every second, ...). Most series are generated continuously but recorded at fixed intervals, so they appear as discrete series.

An important element of the analysis is the **time lag**, the interval between successive observation times. A time series is written

$$\{Z_t : t = 1, 2, 3, \dots\} \qquad \text{or} \qquad Z_1, Z_2, Z_3, \dots$$

The aims of time series analysis fall into two groups.

(i) **Forecasting**: assuming the past pattern persists, analyse the data so far and forecast the future.

(ii) **Understanding and control**: model the system (stochastic process) that generates the series in order to understand or control it.

The first step is always the **time series plot**, with time t on the horizontal axis and Z_t on the vertical axis. It shows the features of the series at a glance and guides the choice of model.

11.1.1 Components of a time series

The traditional approach regards a series as a sum of components and studies each one. A series has a **systematic component** and a **random** (**irregular**) **component**. The systematic component splits further into the **trend component**, which describes long-term change, and the **seasonal** (or cyclical) **component**, which describes repeated short-term change.

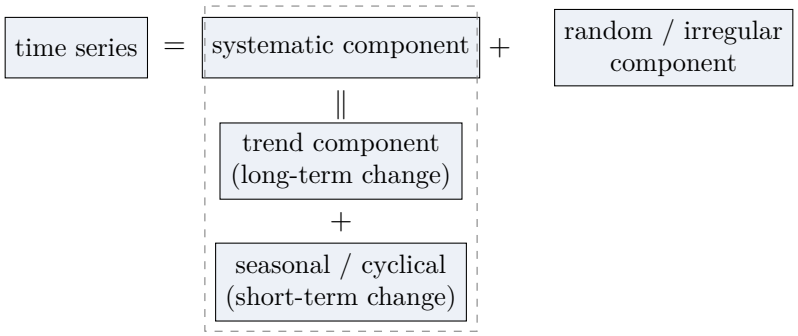


Figure 11.1: Components of a time series.

Definition (Components)

(a) The **random (irregular) component** is variation caused by random factors, unrelated to any regular movement over time. A series made only of a random component around a level is modelled by

$$Z_t = \beta_0 + \epsilon_t. \tag{11.1}$$

(b) The **trend component** is a persistent increase or decrease over time. It need not be linear and may be quadratic or S-shaped. A linear trend gives the **linear trend model**

$$Z_t = \beta_0 + \beta_1 t + \epsilon_t, \tag{11.2}$$

where $\beta_0 + \beta_1 t$ is the trend component and ϵ_t is the irregular component. Curved trends use

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \epsilon_t \quad \text{or} \quad Z_t = \beta_0 + \beta_1 t + \cdots + \beta_k t^k + \epsilon_t.$$

(c) The **seasonal component** is variation that repeats with the week, month or season. Examples are sales of summer goods, rainfall and temperature, which rise in summer and fall in winter.
(d) The **cyclical component** also repeats, but over a long period that is not tied to the seasons. In economics it is the **business cycle**: the Kondratiev cycle (50–60 years), the Juglar cycle (9–10 years) and the Kitchin cycle (about 40 months). A natural example is the roughly 11-year sunspot cycle.

A series whose systematic part is purely seasonal (no trend) can be described with trigonometric or indicator functions. For monthly data with a 12-month seasonal period,

$$Z_t = \beta_0 + \beta_1 \sin\left(\frac{2\pi t}{12}\right) + \beta_2 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.3}$$

The systematic part takes the same value at $t = 1$ and $t = 13$, so it has period 12 months. Many economic series (airline passengers, sales of ice cream or roasted sweet potatoes) have both a trend and seasonality:

$$Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t. \tag{11.4}$$

Here $\beta_0 + \beta_1 t$ is the trend, and the sine and cosine terms form the 12-month seasonal component.

In some series the size of the fluctuations grows over time together with the trend, as in many economic indicators. Then one first takes logarithms and fits a model of type (11.4):

$$\log Z_t = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi t}{12}\right) + \beta_3 \cos\left(\frac{2\pi t}{12}\right) + \epsilon_t.$$

The log transform stabilises the variability and simplifies the model. Forecasts are then on the log scale, so they must be converted back by the **exponential transformation** $\exp(x) = e^x$.

In real data the components are often not clearly separable. A series may also need two different trend lines in different periods, which indicates a **change point**. A main purpose of the time series plot is to reveal such features.

11.2 Methods of time series analysis

Motivation. No single model fits every series. This section surveys the regression-type trend models and the smoothing methods that the rest of the chapter develops.

A traditional approach is to regress the variable on time and analyse the trend. Expressing Z_t as a polynomial trend in t plus an irregular component gives

$$Z_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots + \beta_k t^k + \epsilon_t, \tag{11.5}$$

the **linear polynomial trend model** (or simply **polynomial trend model**). It is a special case of multiple regression,

$$Z_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt} + \epsilon_t,$$

in which the explanatory variables are determined by time: $X_{1t} = t$, $X_{2t} = t^2, \ldots$. If the trend is a nonlinear function of t, we may use a **nonlinear trend model** such as

$$Z_t = \exp(\beta_0 + \beta_1 X_t)\, \epsilon_t.$$

The system generating a series may change over time, so a model that fits well in one period may fit poorly later. Rather than weighting all past observations equally, it is therefore reasonable to give recent observations more weight when forecasting.

If a series has large seasonal swings or much irregular variation, the trend is hard to see. **Smoothing methods** remove the seasonal and irregular components to reveal the overall trend and ease forecasting. The main ones are the **moving average method** and the **exponential smoothing method**.

(a) The moving average gives equal weight to the most recent observations only, not to all observations.

(b) Exponential smoothing gives more weight to recent observations and lets the weights decay exponentially into the past. It is widely used because it adapts flexibly to changes in the generating system and because it is easy to compute without storing much data.

Exponential smoothing is mainly used for forecasting. Moving averages are mainly used to visualise the trend after removing seasonality, which is **seasonal adjustment**. Other approaches include the **decomposition method**, which estimates each component separately and recombines them, and the ARMA/ARIMA models of Box and Jenkins (autoregressive (integrated) moving average). Their three-stage model-building procedure (identification, estimation, diagnosis) is the core of modern time series analysis. Transfer-function models and regression with autocorrelated errors are further options.

This chapter covers trend analysis, moving averages and exponential smoothing. These methods describe the trend in the data. They do not explain why the trend or seasonal variation occurs; they aim only to detect and confirm it. They are empirical, intuitive and convenient, which is why they are widely used.

11.3 Trend analysis

Motivation. When a series fluctuates around a level or follows a clear trend, ordinary least squares on time gives forecasts and prediction intervals. The constant mean model shows the whole workflow in its simplest form.

The choice of model depends on the shape of the series.

(a) A series that moves randomly around a fixed level calls for the constant mean model.

(b) A series with a steady upward or downward trend calls for a linear or a **quadratic trend model**.

(c) A series repeating the same pattern with a fixed period calls for a **seasonal trend model**. If it also has a steady linear trend, we use a **linear seasonal trend model**.

(d) Nonlinear trend models are used mainly for series growing geometrically or moving nonlinearly.

Definition (Three stages of model building)

(i) **Identification**: plot the series, study its features, and choose a tentative model.
(ii) **Estimation**: estimate the parameters by least squares or maximum likelihood.
(iii) **Diagnosis**: analyse the residuals to check the fit. If the fit is inadequate, modify the model or choose another.

The models of trend analysis are fitted exactly as in regression. If the data violate a required assumption (e.g. equal variance), we transform them (e.g. by logs), plot the transformed series again, and choose a tentative model for it.

11.3.1 Constant mean model

A series moving randomly around a fixed level is described by the **constant mean model**

$$Z_t = \beta_0 + \epsilon_t,$$

where the errors ϵ_t are independent with mean 0 and variance σ^2, and β_0 is the unknown constant mean. From the past data $Z_1, \dots, Z_n$ the least squares estimator of β_0 is

$$\hat{\beta}_0 = \frac{1}{n}\sum_{t=1}^{n} Z_t = \overline{Z}.$$

Let $Z_n(l) = E(Z_{n+l} \mid Z_n, Z_{n-1}, \dots)$ be the **l-step-ahead forecast** made at origin n. Under the constant mean model

$$Z_n(l) = \hat{\beta}_0 = \overline{Z} \qquad (l = 1, 2, \dots).$$

Definition (Forecast error, variance estimate and prediction interval)

The mean squared forecast error is

$$E\big[Z_{n+l} - Z_n(l)\big]^2 = \Big(1 + \frac{1}{n}\Big)\sigma^2,$$

and σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=1}^{n}(Z_t - \overline{Z})^2.$$

If the ϵ_t are normal, a $100(1-\alpha)\%$ **prediction interval** for Z_{n+l} is

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1 + \frac{1}{n}\Big)^{1/2},$$

where $t_{\alpha/2}(n-1)$ is the upper $\alpha/2$ point of the t distribution with $n-1$ degrees of freedom.

[Supplement] Why these hold: $Z_{n+l} - \overline{Z} = \epsilon_{n+l} - \bar{\epsilon}$, and ϵ_{n+l} is independent of $\bar{\epsilon} = \frac{1}{n}\sum_{t\le n}\epsilon_t$. So the error has mean 0 and variance $\sigma^2 + \sigma^2/n$. Under normality it is $N\big(0, (1+\frac{1}{n})\sigma^2\big)$ and independent of $\hat{\sigma}^2$, with $(n-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$. Hence $(Z_{n+l} - \overline{Z})/(\hat{\sigma}\sqrt{1+1/n}) \sim t(n-1)$, which gives the interval.

A prediction interval (L, U) for a future *random* value Z_{n+l} satisfies $P(L < Z_{n+l} < U) \ge 95\%$ when $\alpha = 0.05$. It is not a confidence interval for a parameter.

When a new observation Z_{n+1} arrives, the forecasts from origin $n+1$ are

$$\begin{aligned} Z_{n+1}(l) &= \frac{1}{n+1}(Z_1 + \cdots + Z_n + Z_{n+1}) = \frac{n}{n+1}Z_n(l) + \frac{1}{n+1}Z_{n+1} \\ &= Z_n(l) + \frac{1}{n+1}\big[Z_{n+1} - Z_n(l)\big]. \end{aligned}$$

The new forecast is a linear combination of the old forecast $Z_n(l)$ and the forecast error $e_n(l) = Z_{n+1} - Z_n(l)$. It needs only the latest forecast and the latest error, not all past data, so this updating rule saves computation and storage.

Example 11-1 (Statistics course enrolment)

The (hypothetical) numbers of students taking a statistics course each semester are given below.

Sem.	Students	Sem.	Students	Sem.	Students	Sem.	Students
2012/S1	1,372	2015/S1	1,423	2018/S1	1,420	2021/S1	1,435
2012/S2	1,388	2015/S2	1,337	2018/S2	1,406	2021/S2	1,376
2013/S1	1,478	2016/S1	1,366	2019/S1	1,372	2022/S1	1,347
2013/S2	1,404	2016/S2	1,378	2019/S2	1,489	2022/S2	1,389
2014/S1	1,406	2017/S1	1,461	2020/S1	1,425	2023/S1	1,349
2014/S2	1,486	2017/S2	1,418	2020/S2	1,302	2023/S2	1,364

Using the constant mean model, answer the following.
(a) Find the forecast and the 95% prediction interval for enrolment in the first semester of 2026.
(b) If 1,500 students enrol in the first semester of 2024, find the updated forecast.
Solution. Here $n = 24$, $\overline{Z} = 1399.625$ and $\hat{\sigma} = 47.83$. The first semester of 2026 is $l = 5$ steps ahead, and $Z_n(5) = \overline{Z} = 1399.625$. The forecast for 2024/S1 is also 1399.625.
(a) With $t_{0.025}(23) = 2.069$,

$$\overline{Z} \pm t_{\alpha/2}(n-1)\,\hat{\sigma}\Big(1+\frac{1}{n}\Big)^{1/2} = 1399.625 \pm 2.069 \times 47.83 \times \Big(1+\frac{1}{24}\Big)^{1/2} = 1399.625 \pm 100.001.$$

The 95% prediction interval is (1298.624, 1500.626), so roughly 1,300 to 1,500 students are expected.
(b) The forecast error is $Z_{n+1} - Z_n(l) = 1500 - 1399.625 = 100.375$. Hence

$$Z_{n+1}(l) = Z_n(l) + \frac{1}{n+1}\big[Z_{n+1} - Z_n(l)\big] = 1399.625 + \frac{100.375}{25} = 1403.64.$$

11.3.2 Linear trend model

A series following $Z_t = \beta_0 + \beta_1 t + \epsilon_t$ is analysed by the regression methods of Chapter 8.

Illustration 11-1: women in senior professional and technical jobs

Consider a (hypothetical) monthly series, January 2018 to January 2024, of the number of women working in senior professional and technical occupations. Its plot shows a clear linear trend. Fitting the linear trend model gives $R^2 = 0.8373$ and the following ANOVA and coefficient tables.

Source	SS	df	MS	F	p-value
Model	2,170,954	1	2,170,954	365.48	$< 2.2 \times 10^{-16}$
Error	421,736	71	5,940		
Total	2,592,690	72			

Variable	Coefficient	SE	T	p-value
Intercept	−4,578	262.1	−17.46	$< 2 \times 10^{-16}$
Date	0.2689	0.01407	19.12	$< 2 \times 10^{-16}$

The coefficients are significant and R^2 exceeds 80%, so the trend model fits fairly well. The fitted model is

$$\widehat{\text{Female}} = -4578 + 0.269 \times \text{Date},$$

where Date is the number of days since 1 January 1970 (e.g. 1 January 2018 is day 17,532). The residuals must still be checked. Their time plot shows runs of residuals with the same sign over stretches of time, so the residuals are positively correlated, not independent. This violates the regression assumption of independent, identically distributed errors. A regression model with autocorrelated errors would be more appropriate.

11.4 Moving averages

Motivation. Instead of averaging all observations, average only the most recent m. This smooths out local fluctuations, tracks a slowly changing level and, with m equal to the seasonal period, removes seasonality.

Consider the model $Z_t = \beta_0 + \epsilon_t$, where now β_0 may drift slowly over time. Locally the series fluctuates around a level, but the level changes. Instead of the least squares estimator $\hat{\beta}_0 = \overline{Z}$, which weights all observations equally, we use the mean of the most recent m observations:

$$M_n = \frac{1}{m} \sum_{t=n-m+1}^{n} Z_t = \frac{1}{m}(Z_{n-m+1} + Z_{n-m+2} + \cdots + Z_n). \tag{11.6}$$

M_n is the **simple moving average**, and it serves as the forecast of Z_{n+l} for every lead l:

$$Z_n(l) = M_n.$$

The choice of m depends on the purpose. A large m smooths strongly, so M_n reacts slowly to local change. A small m follows local change quickly. To smooth out seasonality, m is set equal to the seasonal period. The method has two drawbacks. The first m observations have no forecast, because the first forecast is for observation $m+1$. And all observations in the window get the same weight regardless of their age.

Moving averages are used less for forecasting than for seasonal adjustment: smoothing out the irregular and seasonal components to reveal the trend. For this, instead of (11.6) one uses the **centered moving average** with $m = 2k + 1$:

$$M_n = \frac{1}{2k+1}(Z_{n-k} + \cdots + Z_{n+k}). \tag{11.7}$$

Illustration 11-2: quarterly births in Korea

The national quarterly numbers of births from 2000/Q1 to 2024/Q1 (97 quarters, Statistics Korea) are, row by row:

178245	152324	151913	157607	161437	136920	131907	129670	140157
123005	119727	114022	133304	121051	117707	122974	132259	115630
115593	113476	118216	106313	108074	106104	119423	110753	111082
110501	126767	117315	128434	124306	127560	111416	115866	111050
120664	107525	110135	106525	120550	111369	117792	120460	128997
116100	116491	109677	128802	117652	123317	114779	119432	105518
109651	101854	116004	107103	110808	101520	117952	110126	108263
102079	112366	102337	102192	89348	98529	89532	89638	80072
89760	82040	80480	74542	83030	75395	73683	70568	73595
68263	68900	61579	70125	66129	66551	57757	68339	60149
64175	56523	64468	56142	56743	52618	60474		

Centered moving averages with $m = 3$ ($k = 1$) and $m = 11$ ($k = 5$) give the following (excerpt of Table 11-3; NA means not computable).

t	Year/Q	Actual	$m = 3$	$m = 11$
1	2000/1	178245	NA	NA
2	2000/2	152324	160827.3	NA
3	2000/3	151913	153948	NA
4	2000/4	157607	156985.7	NA
5	2001/1	161437	151988	NA
6	2001/2	136920	143421.3	143901.1
7	2001/3	131907	132832.3	138062.6
8	2001/4	129670	133911.3	136333.5
9	2002/1	140157	130944	133527.9
10	2002/2	123005	127629.7	129900.6
⋮	⋮	⋮	⋮	⋮
89	2022/1	68339	62081.67	62903.36
90	2022/2	60149	64221	62463.73
91	2022/3	64175	60282.33	60872.18
92	2022/4	56523	61722	60358.09
93	2023/1	64468	59044.33	NA
94	2023/2	56142	59117.67	NA
95	2023/3	56743	55167.67	NA
96	2023/4	52618	56611.67	NA
97	2024/1	60474	NA	NA

A larger m gives a smoother curve, and the overall trend is easier to see. With $m = 11$ the smoothed values change little and the trend stands out, while $m = 3$ still follows short-term ups and downs. The cost is data loss: $m = 3$ loses one value at each end, and $m = 11$ loses five at each end. The unsmoothed series is jagged because of the irregular component, and the smoothed series shows the trend clearly.

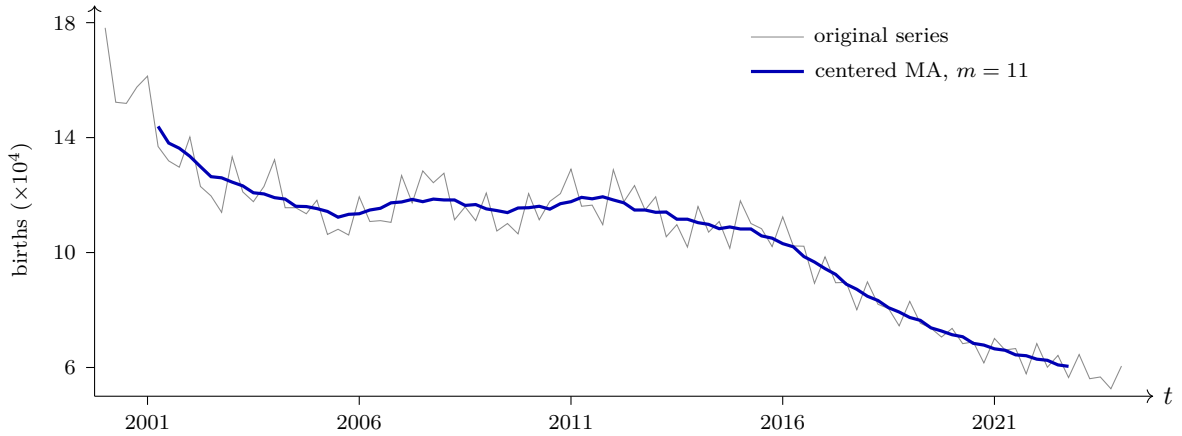


Figure 11.2: Quarterly births and their centered moving average ($m = 11$).

If the series has a linear trend as in (11.2), or a quadratic trend, double or triple moving averages are used instead.

11.5 Exponential smoothing

Motivation. A moving average ignores everything older than m periods and weights the last m equally. Exponential smoothing uses all observations, weights recent ones more, and updates the forecast using just two numbers.

Consider again the model (11.1), whose level may change over time. Given $Z_1, \ldots, Z_n$, exponential

smoothing forecasts Z_{n+1} by weights that decay geometrically into the past:

$$Z_n(1) = c\sum_{t=0}^{n-1}(1-w)^t Z_{n-t} = c\big[Z_n + (1-w)Z_{n-1} + \cdots + (1-w)^{n-1}Z_1\big]. \tag{11.8}$$

The **normalizing constant** c makes the weights sum to 1, i.e. $c\sum_{t=0}^{n-1}(1-w)^t = 1$. As n grows, $c \to w$. So for large n the one-step forecast is

$$Z_n(1) = \sum_{j\geq 0} w(1-w)^j Z_{n-j} = wZ_n + w(1-w)Z_{n-1} + w(1-w)^2 Z_{n-2} + \cdots \tag{11.9}$$

This is a sensible forecast for model (11.1), because it is unbiased:

$$E\big(Z_n(1)\big) = w\sum_{j\geq 0}(1-w)^j E(Z_{n-j}) = w\sum_{j\geq 0}(1-w)^j\beta_0 = \frac{w}{1-(1-w)}\,\beta_0 = \beta_0.$$

Definition (Simple exponential smoothing)

The weight w $(0 < w < 1)$ in (11.8)–(11.9) is the **smoothing constant**, usually between 0.05 and 0.3. The smoothed forecast is the **simple exponentially smoothed value**

$$S_n = Z_n(1),$$

and from origin n every lead has the same forecast: $Z_n(l) = S_n$ for $l = 1, 2, \ldots$. With S_{n-1} the smoothed value at time $n-1$,

$$S_n = wZ_n + (1-w)S_{n-1} = S_{n-1} + w(Z_n - S_{n-1}). \tag{11.10}$$

By (11.10) the new forecast S_n is a linear combination of the previous forecast S_{n-1} and the new observation Z_n. Unlike the moving average, only S_{n-1} and Z_n are needed to produce the updated forecasts $Z_n(l) = S_n$. Computing forecasts recursively by (11.10) is called **simple exponential smoothing**.

11.5.1 Choosing S_0 and w

To start (11.10) we need $S_1 = wZ_1 + (1-w)S_0$, hence an initial value S_0 and a constant w.

(a) *Initial value.* If there is information on data before the sample, it can be used to set S_0; otherwise S_0 must be chosen subjectively. For large n the weight $(1-w)^n$ on S_0 is tiny, so the initial value matters little.

(b) *Smoothing constant.* The choice of w matters a great deal. A small w makes forecasts react slowly to change. A large w makes them follow the latest observations closely. Usually w is treated as fixed over time and taken between 0.05 and 0.3.

A data-based choice uses the **1-step-ahead prediction error** at time $t-1$,

$$e_{t-1}(1) = Z_t - S_{t-1},$$

and picks the w that minimises

$$\mathrm{SSE}(w) = \sum_{t=1}^{n} e_{t-1}^2(1).$$

The choices are linked. If the level of the series changes rapidly, a w close to 1 makes the SSE small, and $S_0 = Z_1$ is appropriate. If the series changes slowly, w is likely to be close to 0, and $S_0 = \overline{Z}$ is a sensible start.

Example 11-2 (KOSPI index)

Monthly closing values of the KOSPI index from January 2018 to December 2024 are given below.

Yr	1	2	3	4	5	6	7	8	9	10	11	12
18	2566.46	2427.36	2445.85	2515.38	2423.01	2326.13	2295.26	2322.88	2343.07	2029.69	2096.86	2041.04
19	2204.85	2195.44	2140.67	2203.59	2041.74	2130.62	2024.55	1967.79	2063.05	2083.48	2087.96	2197.67
20	2119.01	1987.01	1754.64	1947.56	2029.6	2108.33	2249.37	2326.17	2327.89	2267.15	2591.34	2873.47
21	2976.21	3012.95	3061.42	3147.86	3203.92	3296.68	3202.32	3199.27	3068.82	2970.68	2839.01	2977.65
22	2663.34	2699.18	2757.65	2695.05	2685.9	2332.64	2451.5	2472.05	2155.49	2293.61	2472.53	2236.4
23	2425.08	2412.85	2476.86	2501.53	2577.12	2564.28	2632.58	2556.27	2465.07	2277.99	2535.29	2655.28
24	2497.09	2642.36	2746.63	2692.06	2636.52	2797.82	2770.69	2674.31	2593.27	2556.15	2455.91	2399.49

(a) With $S_0 = \overline{Z}$, compute the simple exponentially smoothed values $S_1, \ldots, S_{83}$ for $w = 0.3$ and for $w = 0.1$.
(b) Plot the series with the smoothed values $S_0, \ldots, S_{83}$ for $w = 0.1$ and $w = 0.3$, and describe the effect of w.
Solution.
(a) The series has length $T = 84$, so

$$S_0 = \overline{Z} = \frac{1}{T}\sum_{t=1}^{T} Z_t = 2478.221.$$

For $w = 0.3$, the one-step forecast at $n = 1$, using only Z_1, is

$$S_1 = S_0 + w(Z_1 - S_0) = 2478.221 + 0.3 \times (2566.46 - 2478.221) = 2504.693.$$

Likewise S_2 uses S_1 and Z_2:

$$S_2 = S_1 + w(Z_2 - S_1) = 2504.693 + 0.3 \times (2427.36 - 2504.693) = 2481.493.$$

S_2 uses the data up to time 2 and forecasts Z_3. Continuing gives all of $S_0, \ldots, S_{83}$. Selected values (Table 11-5 lists S_{t-1}, the forecast of Z_t, beside Z_t):

	S_0	S_1	S_2	S_3	$\cdots$	S_{83}
$w = 0.3$	2478.221	2504.693	2481.493	2470.8	$\cdots$	2580.833
$w = 0.1$	2478.221	2487.045	2481.076	2477.554	$\cdots$	2596.527

(b) As with moving averages, the smoothed series moves much more gently than the original (Figure 11.3). With $w = 0.3$ the smoothed series stays fairly close to the original. With $w = 0.1$ it is much smoother and lags further behind the turning points of 2020–2022. [Supplement] For these data the one-step SSE is about 2.75×10^6 for $w = 0.3$ against 6.41×10^6 for $w = 0.1$, so the SSE criterion prefers $w = 0.3$.

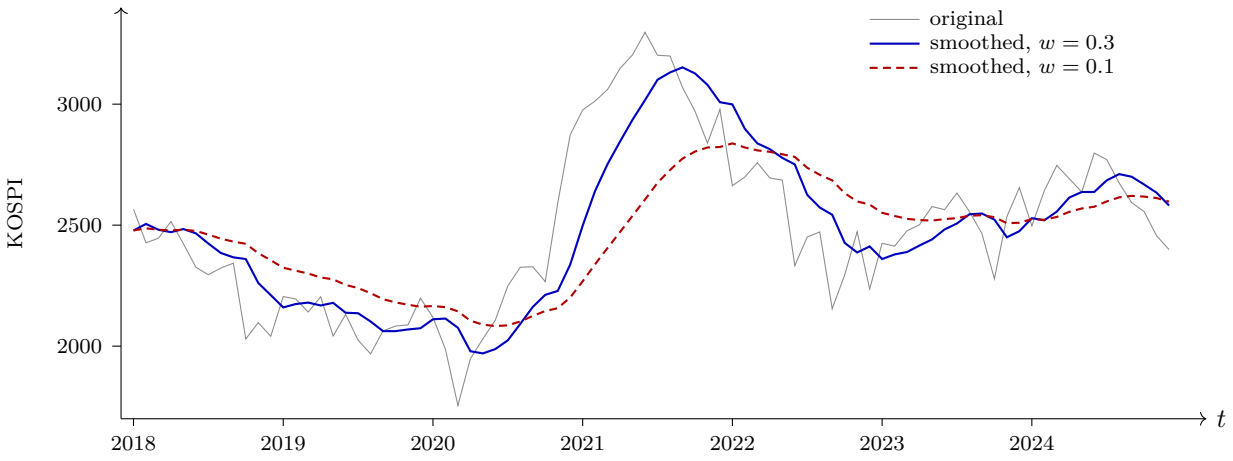


Figure 11.3: KOSPI and its forecasts S_{t-1} by simple exponential smoothing ($w = 0.3$, $w = 0.1$).